The Quantum Threat to Bitcoin

How cryptographically relevant quantum computers could threaten the financial freedom tool that dissidents and human rights defenders rely on — and what the Bitcoin community can do about it.

Human Rights Foundation · Financial Freedom Program · 2025

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EXECUTIVE SUMMARY

A Human Rights Imperative

Bitcoin is a financial lifeline for dissidents resisting authoritarian regimes. The rise of cryptographically relevant quantum computers (CRQCs) with the ability to crack Bitcoin's underlying cryptography could threaten the network's security foundations. A team of Google researchers recently published findings representing a quantum computing breakthrough: an algorithm enabled a quantum computer to carry out operations **13,000 times faster** than a classical supercomputer.

While quantum computing remains largely theoretical, experts suggest CRQCs could emerge within the next five years. CRQCs could put millions of bitcoin stored in early address formats at risk, and endanger the wider trust that underpins Bitcoin. Post-quantum cryptography offers solutions, but migration will require years of technical research, coordination, and global consensus in a decentralized and ideologically divided ecosystem.

The threat has sparked technical, political, and moral debates — including the "burn or steal" dilemma: whether to allow attackers to "steal" funds in early address formats, or "burn" those funds to make them unspendable.

KEY FINDINGS

- **1.72M bitcoin (~$188B)** in early address types, thought dormant or lost, will be highly vulnerable to long-range quantum attacks.
- **4.49M bitcoin (~$495B)** are vulnerable but owners can secure them by moving to quantum-safe addresses.
- Preparing Bitcoin for a post-quantum world is a **human rights imperative**.

Drawing on six months of expert discussions, this report explores CRQC risks for dissidents using Bitcoin for financial freedom. Many thanks to HRF Bitcoin Technical Lead, Alex Li, who drove the primary research.

CONTEXT

Bitcoin as a Human Rights Tool

Bitcoin is a powerful tool for safeguarding human rights, promoting financial freedom, and resisting authoritarian control. Tyrannical governments frequently manipulate and surveil their currencies, seeking to silence critics by confiscating property and freezing bank accounts. Dissidents, journalists, and non-profits under dictatorship face enormous obstacles just to receive donations and pay their bills. HRF's Financial Freedom program supports Bitcoin as a tool for activists facing financial repression: **money that dictators can't stop.**

Bitcoin as resistance: freedom money that dictators can't stop.

Quantum threats to Bitcoin's classical cryptography are still years away. However, if CRQCs emerge before Bitcoin's cryptographic foundations are upgraded, activists who rely on Bitcoin for secure donations, private savings, and uncensored transactions would find their privacy and safety compromised — and dictators might find ways to steal or stop their money.

Making Bitcoin quantum-proof poses political and social challenges. Its decentralization means any upgrade requires coordination and consensus among a diverse, divided user base. Any change to cryptographic signature schemes will require widespread education for users, developers, node operators, and miners. The community must also consider the 1.72 million bitcoin in early addresses most vulnerable to CRQCs.

13,000×

FASTER — GOOGLE'S QUANTUM COMPUTING BREAKTHROUGH VS CLASSICAL SUPERCOMPUTER

5–20 yrs

ESTIMATED TIMELINE BEFORE CRQCS COULD EMERGE, PER EXPERT CONSENSUS

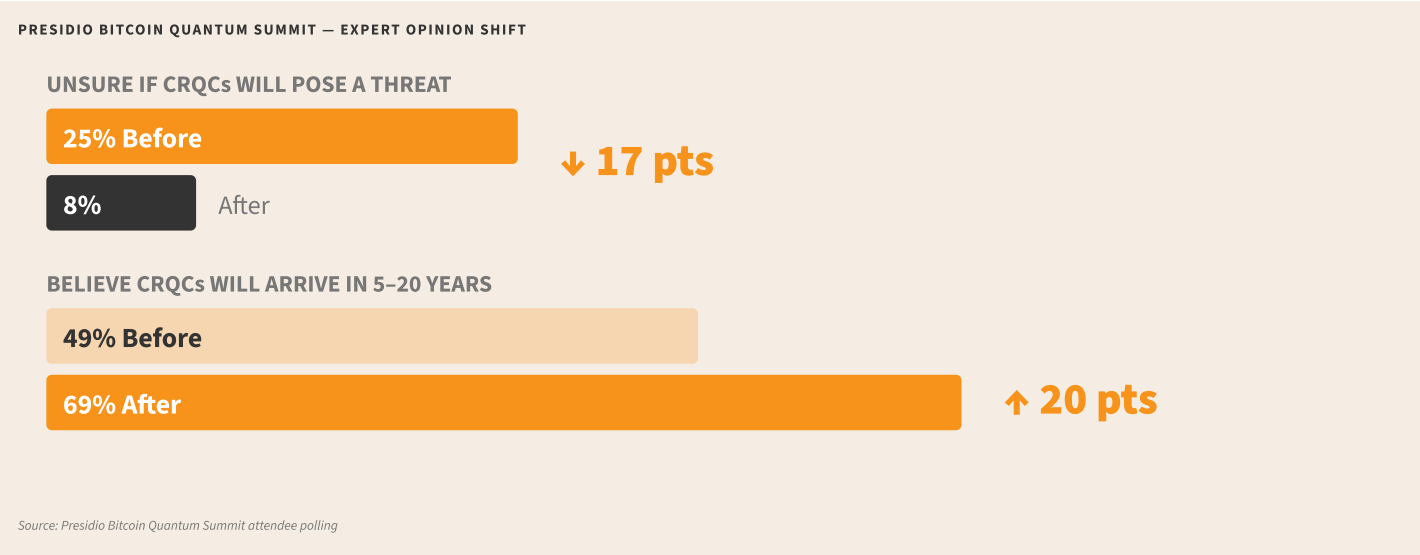
TECHNICAL LANDSCAPE

The Current State of Quantum Computing

Quantum computers exist today, but they remain years away from the scale, stability, and precision required to threaten Bitcoin's cryptography. Even with recent breakthroughs, experts remain divided over whether CRQCs capable of breaking Bitcoin's security will ever emerge. Some emphasize incremental breakthroughs will provide time for solutions; others caution that sudden advances could accelerate the timeline.

At this year's Presidio Bitcoin Quantum Summit — convening quantum physicists, Bitcoin Core developers, cryptographers, wallet engineers, miners, and educators — evolving expert views on CRQCs came into sharp focus.

> *Quantum computing is real. But there are so many things to do. You don't need to worry too much about it, but at the same time, probably it's not a good idea to ignore it... Five to ten years is not a crazy number.*
>
> — SHO SUIGARA, CEO OF BLOCQ



At the outset, 25% of attendees were unsure whether CRQCs would ever pose a meaningful threat. After two days of intensive discussion, uncertainty dropped to just 8%. The share who believed CRQCs would arrive in 5–20 years jumped from 49% to 69%. These shifting perspectives reflect a growing consensus that the threat deserves proactive, serious consideration today.

THREAT ANALYSIS

Bitcoin's Quantum Threat Vectors

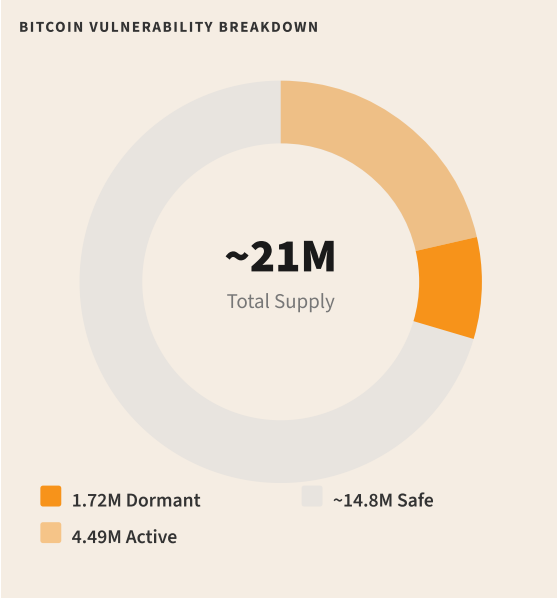
Not all bitcoin will be equally vulnerable. Quantum threats fall into two categories: **long-range** and **short-range attacks**, each exploiting different weaknesses in public key exposure.

Long-Range Attacks

A quantum adversary targets bitcoin whose public keys are already exposed — in older address types, re-used addresses, or taproot (P2TR) addresses. Approximately **6.51 million bitcoin** (over **$718 billion**), nearly a third of total supply, is vulnerable.

Active users can protect funds by moving to quantum-safe addresses. This secures 4.49M of the at-risk bitcoin. But for the remaining **1.72 million bitcoin** ($188B) — owners' private keys thought to be lost — no migration can occur. Satoshi's estimated 1.1M BTC ($130B+) sits in P2PK addresses with exposed public keys.

This raises the **"burn or steal"** debate: render vulnerable coins unspendable, or leave them exposed to quantum theft? It cuts to the heart of Bitcoin's values — decentralization, sovereignty, immutability, and property rights.



Short-Range Attacks

Short-range attacks exploit the brief window between broadcast and confirmation of a transaction. A CRQC could intercept an unconfirmed transaction, derive the private key from the exposed public key in real time, and redirect funds. All Bitcoin is vulnerable to short-range attacks until post-quantum signatures are adopted.

> *Someone is going to transition through this phase transition, and it's going to come just as fast as AI came and hit people in the face.*
> — TERRY RUDOLPH, CEO OF PSIQUANTUM

VULNERABILITIES

Infrastructural Risks

Modern Bitcoin infrastructure introduces another underappreciated vulnerability. Most wallets, companion software, and portfolio trackers store public keys to calculate balances and generate wallets. If these companies are hacked in a CRQC world, attackers could steal users' funds. Hardware wallets must adapt to slower computations and bulkier keys. Accounting platforms, custody providers, and financial institutions will need to retool. Multisig coordination, watch-only setups, and automated workflows will all need reevaluation.

SOLUTIONS

Preparing Bitcoin for the Quantum Era

Quantum-Resistant Signature Schemes

Integrating quantum-resistant signature schemes represents the only durable solution. Two proven approaches exist:

LATTICE-BASED

More compact; supports multisig, key aggregation, deterministic derivation — features useful for human rights defenders. Introduces new cryptographic assumptions requiring careful vetting.

10×

larger than current signatures

Examples: CRYSTALS-Dilithium 44, FALCON 512

HASH-BASED

Most mature post-quantum option with proven foundations. Larger signatures make key aggregation and multisig more complex to implement.

38×

larger than current signatures

Examples: SPHINCS+, XMSS, Lamport

These dramatic size increases would reduce transactions per block, decrease throughput, and increase storage demands on full nodes. Accommodating larger signatures is likely to divide the community.

BIP 360

Bitcoin Improvement Proposal 360 is **signature scheme-agnostic** — it makes taproot addresses more quantum-resistant and provides a flexible framework for post-quantum algorithms.

IMPLEMENTATION

Upgrades to Bitcoin Education & Design

Upgrading Bitcoin is as much a human challenge as a cryptographic one. Any successful soft fork will necessitate user education, thoughtful UI design, and coordination across users, developers, hardware manufacturers, node operators, and civil society.

Wallets are tightly bound to the current elliptic curve model. Quantum-resistant algorithms would introduce much larger signatures, slower signing, and complex verification paths — fundamentally changing how wallets operate. Hardware wallets must adapt to slower computations and bulkier keys while preserving seamless experience.

Many Bitcoin users remain unaware their coins may be vulnerable to long-range attacks from public key exposure. Encouraging migration to quantum-safe outputs, when the threat remains abstract, will be difficult.

> *Sufficiently powerful quantum computers are hypothetical, and if they happen, there will likely be a long series of incremental breakthroughs that give us time for more fundamental solutions.*
> — PIETER WUILLE, BITCOIN CORE CONTRIBUTOR

Accounting platforms, custody providers, and financial institutions will retool their systems. Developers face balancing complexity with usability. For Bitcoin to remain a reliable human rights tool in the quantum era, upgrades must be **inclusive, accessible, and resilient**.

GOVERNANCE

The Challenge of Decentralized Consensus

Bitcoin has no central authority to enforce upgrades. Every soft fork depends on voluntary adoption and grassroots coordination. Signature schemes that increase transaction sizes 10× or more will trigger debates about block space, throughput, and scalability. Previous improvements like SegWit took years to reach adoption. Convincing a global user base to prepare for a still-hypothetical quantum future will be even harder.

WHAT SUCCESS REQUIRES

A quantum-resilient soft fork must be **socially durable** as well as technically correct. Upgrades must provide clear benefits, usable defaults, and migration tools. Consensus is slow, fragile, and precious. Above all, a soft fork must remain faithful to Bitcoin's principles: **privacy, decentralization, and freedom from coercion**.

THE DILEMMA

Burn, Steal, Hourglass

To protect 1.72 million dormant bitcoin from long-range attack, some advocate a **"burn"** — rendering quantum-vulnerable coins unspendable after a migration window. Advocates argue this protects monetary integrity, prevents destabilizing wealth redistribution, and reinforces that cryptographic theft is not valid ownership.

Others worry this violates property rights and Bitcoin's promise of immutability. The theft of millions of bitcoin could undermine value for all holders — but so could a unilateral decision to freeze coins. This debate cuts to the heart of what Bitcoin means.

🔥

"BURN"
Render vulnerable coins unspendable

⚖️

DO NOTHING
Risk quantum seizure of dormant coins

CONCLUSION

The Path Forward

The quantum threat to Bitcoin is not imminent — but it is real. The shifting expert consensus at the Presidio summit, where uncertainty dropped from 25% to 8%, signals that preparation must begin now. The stakes are uniquely high for human rights defenders — Bitcoin is often their only tool for receiving funds, preserving savings, and transacting freely under authoritarian surveillance.

The solutions exist. BIP 360 provides a flexible, scheme-agnostic framework. But implementation demands social consensus in a decentralized ecosystem where every change is debated and voluntarily adopted.

> *Above all, a soft fork must remain faithful to Bitcoin's underlying principles: privacy, decentralization, and freedom from coercion.*
>
> **Anything less risks the financial freedom of the dissidents and human rights defenders who need Bitcoin the most.**

TECHNICAL REFERENCE

Quantum-Resistant Signature Comparison

PROPERTY	LATTICE-BASED	HASH-BASED
Examples	CRYSTALS-Dilithium 44, FALCON 512	SPHINCS+, XMSS, Lamport
Signature Size	~10× current	~38× current
Maturity	Newer; needs vetting	Most mature PQ option
Multisig	Easier	More complex
Key Aggregation	Supported	Complex
Human Rights Use	Collaborative custody	High-value storage

Source: Chaincode Labs — Bitcoin and Quantum Computing: Current Status and Future Directions

6.51M
BITCOIN VULNERABLE TO LONG-RANGE QUANTUM ATTACKS

$718B+
TOTAL VALUE AT RISK

1.72M
DORMANT — NO MIGRATION POSSIBLE

The Human Rights Foundation promotes and protects human rights globally. HRF's Financial Freedom program supports Bitcoin as a tool for activists facing financial repression.

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